Simulation of a Stochastic Penetration Process: Bernoulli and Rademacher Models

Homework in Probability

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Abstract

We study sums of independent and identically distributed increments in two cases: Bernoulli and Rademacher. We compare the total, the sample average, and the normalized total, focusing on the Law of Large Numbers (LLN), variance, and the Central Limit Theorem (CLT). Since the number of attackers m does not appear in the original definitions of the three statistics, we state an operational assumption that connects m to the model and distinguish the penetration of distinct systems from the sum of attackers' contributions.

1 Model and notation

Let n be the number of systems (or time steps), and let $X_1, \dots, X_n$ be a sequence of i.i.d. random variables. The cumulative score after n steps is

$$S_n = \sum_{i=1}^{n} X_i.$$

The three quantities of interest are

$$S_n, \qquad \overline{X}_n = \frac{S_n}{n}, \qquad Z_n = \frac{S_n}{\sqrt{n}}.$$

The parameter m represents the number of attackers. To incorporate it into the model, we must specify how multiple attackers contribute to the score. We adopt two interpretations, consistent with the purposes of the two models but not identical to each other.

2 Bernoulli model: systems penetrated at least once

Suppose that each attacker attempts to compromise each system with probability q, independently of the other attackers and other systems. Define $X_i = 1$ if system i is compromised by at least one of the m attackers, and $X_i = 0$ otherwise. The effective penetration probability is therefore

$$p_m = \mathbb{P}(X_i = 1) = 1 - (1 - q)^m.$$

It follows that $X_i \sim \text{Bernoulli}(p_m)$ and

$$\mathbb{E}[X_i] = p_m, \qquad \text{Var}(X_i) = p_m(1 - p_m).$$

The sum S_n counts the number of distinct systems compromised at least once, so

$$S_n \sim \text{Binomial}(n, p_m), \quad \mathbb{E}[S_n] = np_m, \quad \text{Var}(S_n) = np_m(1 - p_m).$$

The mean and normalized total have, respectively,

$$\mathbb{E}\left[\frac{S_n}{n}\right] = p_m, \qquad \mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{p_m(1-p_m)}{n},$$
$$\mathbb{E}\left[\frac{S_n}{\sqrt{n}}\right] = p_m\sqrt{n}, \qquad \mathrm{Var}\left(\frac{S_n}{\sqrt{n}}\right) = p_m(1-p_m).$$

2.1 Law of Large Numbers

If p_m remains fixed as $n \to \infty$, the Law of Large Numbers implies

$$\frac{S_n}{n} \xrightarrow{\mathbb{P}} p_m.$$

The same result follows from Chebyshev's inequality: for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - p_m\right| \geq \varepsilon\right) \leq \frac{p_m(1-p_m)}{n\varepsilon^2} \longrightarrow 0.$$

Thus, the average does not generally tend to zero: it tends to the effective penetration probability p_m. Moreover, for $q > 0$ and $m \geq 1$, we have $p_m > 0$; in this case, $S_n/\sqrt{n}$ grows like $p_m\sqrt{n}$ and diverges to $+\infty$ in probability. Therefore, the uncentered statistic $S_n/\sqrt{n}$ does not converge to a standard normal distribution in the Bernoulli model.

To obtain a normal limit, the sum must be centered and scaled by its standard deviation:

$$\frac{S_n - np_m}{\sqrt{np_m(1-p_m)}} \xrightarrow{d} \mathcal{N}(0,1), \qquad 0 < p_m < 1,$$

which is the standard form of the CLT for i.i.d. Bernoulli random variables.

3 Rademacher model: symmetric contributions

In the Rademacher model, each increment satisfies

$$\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = \frac{1}{2}.$$

Therefore,

$$\mathbb{E}[X_i] = 0, \qquad \mathrm{Var}(X_i) = 1,$$

and, by independence,

$$\mathbb{E}[S_n] = 0, \qquad \mathrm{Var}(S_n) = n.$$

For the other two statistics, we obtain

$$\mathbb{E}\left[\frac{S_n}{n}\right] = 0, \qquad \mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{1}{n},$$
$$\mathbb{E}\left[\frac{S_n}{\sqrt{n}}\right] = 0, \qquad \mathrm{Var}\left(\frac{S_n}{\sqrt{n}}\right) = 1.$$

3.1 Law of Large Numbers

The LLN states that

$$\frac{S_n}{n} \xrightarrow{\mathbb{P}} 0.$$

Indeed, by Chebyshev's inequality,

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| \geq \varepsilon\right) \leq \frac{1}{n\varepsilon^2} \longrightarrow 0.$$

The total S_n, however, does not concentrate around a constant: its variance grows linearly with n, and its standard deviation grows like $\sqrt{n}$. Dividing by $\sqrt{n}$ keeps the variance equal to one, yielding a non-degenerate scale.

3.2 Central Limit Theorem

Since the increments are i.i.d. with zero mean and finite, positive variance, the CLT gives

$$\boxed{\frac{S_n}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).}$$

For finite n, $S_n/\sqrt{n}$ still takes values in a discrete set; as n increases, its discrete distribution progressively approximates the standard normal curve. Convergence in distribution does not mean that every trajectory converges to a particular curve: it concerns the distribution of values over many replications.

4 Summary of means and variances

Writing $p = p_m$ for the Bernoulli model, we have:

Quantity	Bernoulli	Rademacher	Interpretation
S_n	$\mathbb{E} = np$, $\mathrm{Var} = np(1-p)$	$\mathbb{E} = 0$, $\mathrm{Var} = n$	total
S_n/n	$\mathbb{E} = p$, $\mathrm{Var} = p(1-p)/n$	$\mathbb{E} = 0$, $\mathrm{Var} = 1/n$	average
$S_n/\sqrt{n}$	$\mathbb{E} = p\sqrt{n}$, $\mathrm{Var} = p(1-p)$	$\mathbb{E} = 0$, $\mathrm{Var} = 1$	CLT scaling only after centering in the Bernoulli c

5 Interpreting the number of attackers m

5.1 Bernoulli: system coverage

The aggregate score counts distinct systems compromised. For fixed q,

$$p_m = 1 - (1-q)^m$$

is increasing in m and tends to one as m increases. Consequently, $\mathbb{E}[S_n] = np_m$ increases with m: more independent attackers increase the probability that each system is compromised. For very large m, the total approaches n; it cannot exceed n, because each system is counted only once.

5.2 Rademacher: sum of contributions from multiple attackers

To preserve the original Rademacher definition, S_n is initially the score of a single attacker. For m independent attackers, introduce copies $X_i^{(j)}$ and the aggregate total

$$T_{n,m} = \sum_{j=1}^{m} \sum_{i=1}^{n} X_i^{(j)}.$$

Then

$$\mathbb{E}[T_{n,m}] = 0, \qquad \mathrm{Var}(T_{n,m}) = mn.$$

The average per contribution and the normalized total satisfy

$$\frac{T_{n,m}}{mn} \xrightarrow{\mathbb{P}} 0, \qquad \frac{T_{n,m}}{\sqrt{mn}} \xrightarrow{d} \mathcal{N}(0,1),$$

when $mn \to \infty$. Increasing m does not shift the mean of the total, which remains zero, but it increases the absolute variability of the aggregate total. The parameter m does not change the distribution of a single attacker's score; it changes the number of contributions being summed.

Remark 1. *The two models represent different notions of an aggregate score. In the Bernoulli model, we count systems compromised at least once; in the Rademacher model, we sum positive and negative contributions. They are useful for studying two distinct probabilistic regimes, but should not be interpreted as the same operational metric without further assumptions.*

6 Simulation procedure

The accompanying HTML page runs Monte Carlo simulations. The user chooses n, m, the number of replications, and, in the Bernoulli case, the probability q. For each replication, the total, the average, and the normalized total are computed. The page also displays cumulative trajectories, histograms, and empirical means and variances alongside their theoretical values.

For the Bernoulli model, each simulated increment is a Bernoulli random variable with parameter p_m. For the Rademacher model, m independent copies of a single attacker's trajectory are simulated; the histograms of the three requested statistics combine results at the individual-attacker level, while the total across all attackers is computed separately as $T_{n,m}$. This keeps the meaning of the statistics consistent with their mathematical definitions.

7 Expected results and conclusions

As n increases, the histogram of the average S_n/n concentrates around p_m in the Bernoulli case and around zero in the Rademacher case. The variances of the averages decrease as $p_m(1 - p_m)/n$ and $1/n$, respectively. The unnormalized total, on the other hand, exhibits increasing dispersion. In the Rademacher case, the histogram of $S_n/\sqrt{n}$ approaches the standard normal distribution, in accordance with the CLT; in the Bernoulli case, the normal limit requires centering by np_m and standardization by $\sqrt{np_m(1 - p_m)}$.

The role of m depends on the operational definition of the score. If the score counts systems compromised at least once, more attackers increase the effective penetration probability. If attackers' contributions are summed with signs, as in the Rademacher model, the mean remains zero while the variance of the aggregate grows linearly with m. Distinguishing the total, the average, and the normalized total makes these behaviors visible at different scales.